

ANNEX B

NJC H1 Maths Preliminary Examination

Qn/No	Topic Set	Answers
1	Quadratic Equations Integration Techniques, Definite Integrals	$-0.5 < m < 4.5$
2	Differentiation (Stationary points)	(ii) V is maximum when $x = \frac{5 - \sqrt{7}}{6}$ (iii) $\frac{5 - \sqrt{7}}{6} < x < 1$
3	Differentiation (Gradients, Tangents and Normals) Simultaneous Equations	(i) $y = 1.2586x - 3.2399$ (ii) 1.4568
4	Differentiation, Integration	(a) $\frac{1}{\sqrt{x}} - \frac{60x^3}{5x^4 - 1}$ (b) $6\ln x - 3 - 4\ln 2x + 1 + C$ (c) $-4xe^{1-2x^2}, -2e^{1-2x^2} + \frac{3x^{\frac{2}{3}}}{2} + C$
5	Graphs Quadratic Equations and Inequalities, Integration (Area)	(i) $y = 7, x = 2, (0, 3), (6/7, 0)$ (ii) $x = -\frac{2}{7}$ or $\frac{5}{2}$ or 2.5, $-\frac{2}{7} \leq x < 2$ or $x \geq 2.5$ (iii) 4.585
6	Sampling Methods	
7	Probability	(i) $\frac{5}{12}$ (ii) $\frac{2}{9}$ (iii) $\frac{17}{20}$
8	Probability	(a) (i) $\frac{1}{8}$ (ii) $\frac{3}{40}$ (iii) 84 (b) 0.4, No
9	Binomial Distribution, Normal Approximation to Binomial Distribution	(i) 0.0911 (ii) 0.658 (iii) 0.479 0.996
10	Hypothesis Testing, Sampling Theory	(i) 299.6, 6.70 (ii) p-value = 0.042036357

		(iv) No, by Central Limit Theorem as n is large (v) p-value = 2×0.0420
11	Normal Distribution	(ii) 0.0228 (iii) 0.787 (iv) 0.0317 14.64
12	Correlation & Regression	(8.78,168)